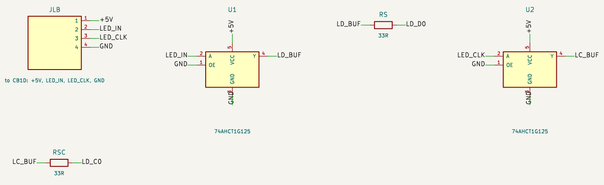
PF233-LB1: LB1-v6 display: 66 x SK9822-EC20 clocked chain with 3V3 -> 5 V data and clock buffers

How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.
Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

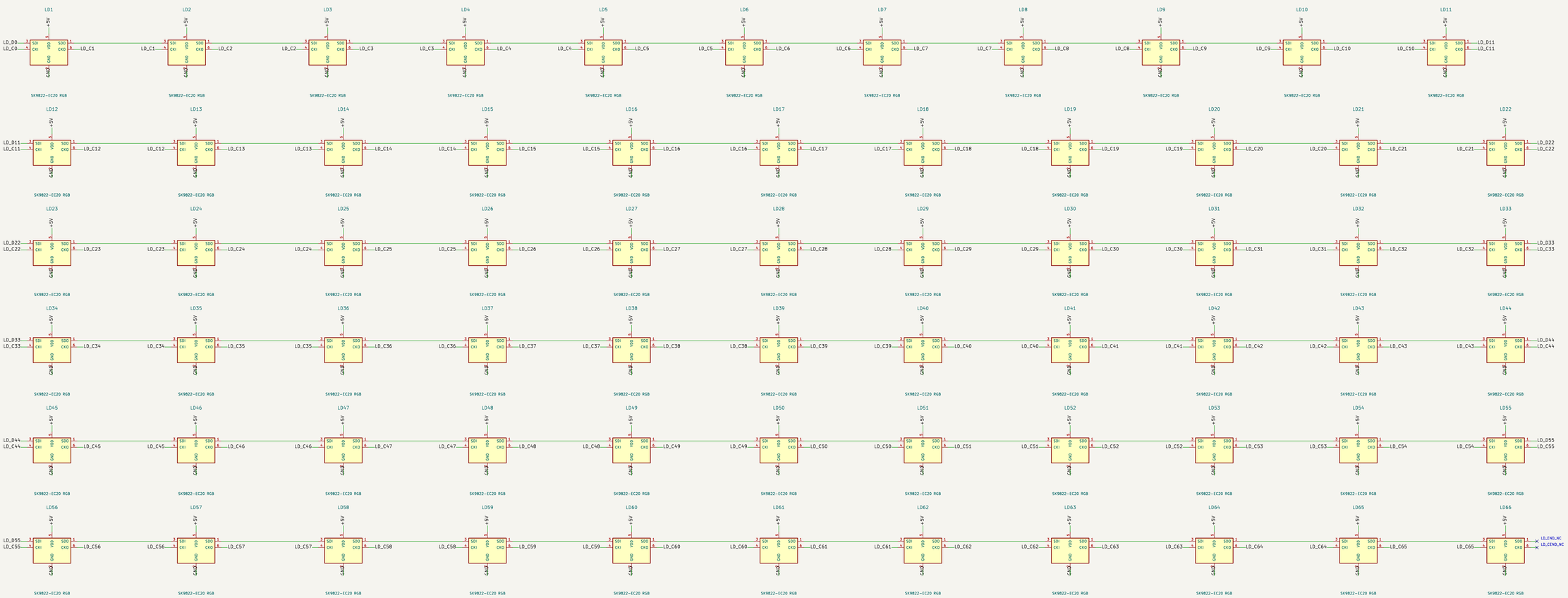
1 Data and clock entry: 3V3 -> 5 V buffers

JLB: +5V, LED_IN, LED_CLK, GND from CB1D. U1 / U2 (74AHCT1G125) buffer LED_IN / LED_CLK to 5 V logic:
RS / RSC 33 R series damping -> LD_D0 / LD_C0.



2 SK9822-EC20 clocked chain, 66 pixels

SDO of each pixel goes straight to SDI of the next (wired): CKO -> CKI of the next by net label (LD_Cn).
Pins: 1 SDO, 2 GND, 3 SDI, 4 CKI, 5 VDD (+5V), 6 CKO. LD66 SDO / CKO are left open (LD_END_NC / LD_CEND_NC).



3 Decoupling

CL1 to CL11: 100 nF across +5V / GND; CB 10 uF bulk.



Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
LB1-v6 display: 66 x SK9822-EC20 clocked chain with 3V3 -> 5 V data and clock buffers
ARX / PHI-FOLD 233
Sheets /
File: PF233-LB1_functional.kicad_sch

Title: PF233-LB1 functional schematic

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KiCad E.D.A. 9.0.9		Id: 1/1